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List of Theorems and Definitions

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This is just a summary of the definitions and theorems used. The theorems are paraphrased, and not the actual theorems stated. This is a growing list.

1 Monads

1.1 Monads and their Algebras

Note on composeability of functors and natural transformations

Definition 1.1.1. Monad

Definition 1.1.2. T-algebra

η and μ of a monad T are referred to as the identity and multiplication of a monad

Definition 1.1.3. Morphism of T-algebras

Definition 1.1.4. Eilenberg-Moore category of T, written $\mathbb{X}^T$

Theorem 1.1.5. *Every adjunction determines a monad. Proof included.*

Definition 1.1.6. Monad defined by the adjunction

Theorem 1.1.7. *Every monad is defined by its T-algebras: every monad determines an adjunction and the monad defined by this adjunction is the same as the first monad. Proof included, with full description of the adjunction.*

Theorem 1.1.8. *Comparison of adjunctions with algebras: existence of Kleisli comparison functor. Proof included, full description of functor.*

Theorem 1.1.9. *Existence of adjunction between first category $\mathbb{A}$ and $\mathbb{X}^T$. Proof included, with full definition of adjunction.*

Definition 1.1.10. Comparison functor

Definition 1.1.11. Monadic functor

Theorem 1.1.12. *Kleisli category of a monad. Proof included.*

Theorem 1.1.13. *Comparison theorem for the Kleisli construction.*

Theorem 1.1.14. *The Eilenberg-Moore and Kleisli construction as terminal and initial objects.*

1.2 Beck's Theorem

Definition 1.2.1. Coequaliser, fork, split coequaliser, split fork, absolute coequaliser, reflexive coequaliser

Theorem 1.2.2. *Beck's theorem characterising algebras: the 3 equivalent conditions about K being an equivalence, and how the right adjoint creates coequalisers. Proof included.*

Definition 1.2.3. A functor reflects isos

Theorem 1.2.4. *Beck's monadicity theorem: 4 equivalent conditions of monadicity, in terms of preserving coeqaliser diagrams of a given form. Proof included.*

Theorem 1.2.5. *Crude monadicity theorem: in which a functor is monadic if it has a left adjoint, reflects isos and preserves reflexive coequalisers. Proof included*

1.3 Comonads and Precomonadicity

Definition 1.3.1. Comonad

Definition 1.3.2. Precomonadic